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Entropy-Based Stochastic Decision Support for Inventory Control in Finite-Capacity Warehouses

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Abstract: Efficient inventory control in finite-capacity warehouses requires a quantitative understanding of the stochastic interactions between supply, demand, and storage limitations. An entropy-based stochastic decision-support framework was developed to characterize inventory dynamics and evaluate operational uncertainty under finite-capacity constraints. The warehouse was modeled as a finite-capacity birth-death process in which inventory replenishment and order fulfillment were governed by stochastic supply and demand rates, respectively, while the supply-to-demand ratio is defined as the warehouse utilization factor. The steady-state probability distribution of inventory levels was derived, and Shannon entropy was employed to quantify the uncertainty associated with warehouse occupancy. It was demonstrated that entropy reached its maximum when the warehouse utilization factor equaled unity, indicating that all inventory states become equally probable and that the inventory system operates under the greatest level of stochastic uncertainty. This critical operating point also represents the transition between demand-driven and supply-driven regimes. When the utilization factor was less than one, warehouse performance was primarily constrained by stockout risk, whereas values exceeding one led to increasing blocking probability caused by storage saturation. The trade-off between blocking probability and stockout probability was further investigated, revealing that their combined probability is minimized at the critical utilization condition. Consequently, the complementary probability of the warehouse operating in standard inventory states was maximized. In addition, the stochastic race condition between stockout and blocking events was analyzed to determine which operational constraint is expected to occur first under varying utilization levels. The proposed framework provides both theoretical insight and practical decision support for inventory planning, warehouse capacity management, and procurement policy optimization by establishing a direct relationship between stochastic inventory behavior, information entropy, and operational performance in finite-capacity warehouse systems.

Keywords: Stock inventory; Information and entropy; Utilization; Warehouse; Stockout probability; Blocking probability

1 Introduction

In the modern global supply chain, the warehouse is no longer just a passive storage hub; it is a dynamic center directly tied to an organization's competitive advantage [1]. Effective stock inventory management sits at the intersection of operational efficiency and customer satisfaction [2]. As businesses scale and consumer demands for rapid, accurate fulfillment intensify, the pressure on warehouse ecosystems to operate flawlessly has reached unprecedented heights [3]. Managing a warehouse inventory involves a delicate balancing act. Maintaining excess stock ties up valuable working capital and increases holding costs, while under-stocking leads to stockouts, backorders, and severely damaged customer relationships [4]. This equilibrium is further complicated by: (i) demand volatility—unpredictable market fluctuations that disrupt traditional forecasting; (ii) spatial constraints—the physical limitations of maximizing storage density without compromising retrieval speed; (iii) human error—inefficiencies born from manual tracking and disorganized picking routes.

In modern smart warehouses, managers do not just observe stock; they use algorithmic frameworks to automate decisions, predict vulnerabilities, and balance trade-offs [5]. Framing this around intelligent management decisions means focusing on how an enterprise can use a stochastic framework to make automated, proactive choices rather than reactive ones. The integration of autonomous systems within logistics has shifted from isolated robotic deployments to unified, goal-directed orchestration [6, 7]. While recent studies have demonstrated the efficacy of machine learning in optimizing warehouse throughput [8, 9] and facilitating human-robot collaboration [10, 11], these implementations frequently rely on heuristics that lack sufficient interpretability and formal stability guarantees. In industrial environments characterized by high volatility, there is a critical requirement for analytical governance that can bridge the gap between raw Internet of Things telemetry and high-level autonomous strategy [12, 13].

This study addresses this gap by introducing an analytical decision engine grounded in stochastic performance modeling. Unlike purely data-driven predictive models, the proposed approach utilizes a birth-death process framework to derive prescriptive insights. This analytical foundation provides a verifiable methodology for system orchestration, ensuring that automated adjustments remain stable under fluctuating demand conditions [14, 15]. Consequently, the proposed model serves as the computational core for smart logistics centers, transforming reactive warehouse execution systems into proactive, performance-optimized environments. Intelligent management relies on minimizing chaos and uncertainty. By calculating the entropy of the stock steady-state probability, a facility manager can quantify the uncertainty or inconsistency of the inventory level [16]. High entropy means the warehouse is wildly uncertain, and low entropy means it stays consistently empty or full, signaling a need for structural adjustment. In general, this system can be framed as an M/M/1/M model (where the current stock inventory level N obeys a truncated geometric distribution of size M, with $0 \leq N \leq M$) being a mathematical engine inside an intelligent decision support system [17]. In a warehouse, supply items arrive at a constant rate λ, and demand orders are processed at a constant rate μ; then, $\rho = \lambda/\mu$ represents the traffic intensity, or utilization factor, of the warehouse.

This study develops a closed-form stochastic framework to analyze the trade-offs between procurement blocking probability p_M (when the warehouse is full, i.e., $N = M$) and stockout probability p_0 (when the warehouse is empty, i.e., $N = 0$). The intelligent decision should balance the trade-off between minimizing stockout risk and avoiding overcapacity. In this direction, it is shown that, at the maximum uncertainty point of the system utilization (i.e., traffic intensity) parameter, which also represents the transition point between supply-driven and demand-driven operating regimes, their sum $p_0 + p_M$ has a minimum value; that is, the probability of being in any other state (i.e., so-called standard working probability), as a complement, is $p_{\text{work}} = 1 - (p_0 + p_M)$ and has a maximum value. Furthermore, this study evaluates which of these two probabilities is larger in an inventory system, i.e., the probability that a stockout occurs before blocking, and conversely, the probability that blocking occurs before a stockout. All these measures are investigated with respect to the system utilization parameter ρ and system size M.

An intelligent system uses the model's closed-form equations to instantly calculate the ideal operating point based on live data feeds: (i) the system continuously monitors the supply rate λ and demand rate μ. If the ratio ρ shifts drastically, the system uses the formulas to predict a stockout, or an overflow, before it physically happens; (ii) by mapping penalty costs to stockout probability p_0 and blocking probability p_M, management algorithms can execute optimal pricing, or procurement decisions, autonomously. Therefore, this study focuses on using these mathematical insights to help managers (or automated systems) make better structural and operational choices under uncertainty.

In this work, depending on the concepts of information i and entropy $S = E(i)$ associated with a supply-demand warehouse, with a current stock inventory level N, and a maximum stock size M, and modeling the warehouse by a birth-death process of size $M + 1$, and states n, $n = 0, 1, 2, \ldots, M$, this study promotes a stochastic analysis of the supply-demand process dynamics. The information i possessed by a warehouse (i.e., carried by the random variable N) is an essential characteristic of the distinct states of the system (i.e., of the birth-death process) and, by definition, the entropy $S = E(i)$ of the warehouse is the uncertainty associated with the warehouse. This study adopts the entropy formula [18] and its formulation in terms of random variables, proposed initially by Shannon [19] (focusing only on the discrete case).

The remainder of this study is structured as follows. Section 2 reports the basic model and introduces the measures for managing the stock warehouse. Section 3 assesses the probability that a stockout occurs before/after a procurement blocking. Section 4 provides the conclusion of this study.

2 Model Description

This study models a warehouse supply and demand system as a birth-death process of finite size, which is an elegant way to capture inventory dynamics under capacity constraints. In this framework, the warehouse has a strict maximum capacity M. The state n, $n \in \{0, 1, \ldots, M\}$ of the birth-death process represents the current stock inventory level N, where $0 \leq N \leq M$, i.e., the number of items currently in stock.

Because it is a structure with one server and finite capacity, this study defines the birth-death transition rates as follows:

• Supply channel (births): Items arrive at a constant rate λ, but the warehouse cannot exceed its maximum capacity M. Therefore, the birth rate drops to zero once the warehouse is full.

• Demand channel (deaths): Orders arrive and are processed at a constant rate μ. However, if there is no inventory ($n = 0$), no demand can be satisfied.

Therefore, this warehouse system can be modeled as an open queueing system in a steady state. If items arrive at rate λ, and orders are processed at rate μ, then $\rho = \lambda/\mu$ represents the traffic intensity, or utilization factor, of the warehouse. The quantity N, which determines the specific states of the system, represents a random variable with probability distribution $p_n = \text{Prob}\{N = n\}$, $n = 0, 1, \ldots, M$. Therefore, p_n is the steady-state probability that the warehouse contains exactly n items, and is governed by the probability continuity law (via the global balance equations) and probability conservation law (via the normalization factor) [20–23]. Therefore, the following can be derived [24]:

$$\begin{aligned} &\frac{p_n(\rho)}{p_0(\rho)} = \prod_{j=0}^{n-1} \frac{\lambda_j}{\mu_{j+1}} = a_n \cdot \rho^n, \quad a_n > 0, \quad n = 0, 1, \ldots, M \\ &p_0(\rho) = \frac{1}{f_M(\rho)}, \quad f_M(\rho) = \sum_{n=0}^{M} a_n \cdot \rho^n, \quad \rho > 0 \end{aligned} \tag{1}$$

Considering an M/M/1/M queueing system, the state-dependent birth rate λ_n, $n = 0, 1, \ldots, M-1$, (representing the arrival of new stock, when the inventory level is n items) and death rate μ_n, $n = 1, 2, \ldots, M$, (representing the fulfillment of an order, when the inventory level is n items) are defined as:

$$\lambda_n = \begin{cases} \lambda, & 0 \le n < M \\ 0, & n \ge M \end{cases} \qquad \mu_n = \begin{cases} \mu, & 0 < n \le M \\ 0, & n = 0 \end{cases} \tag{2}$$

Thus, the probability of having N items in stock (or inventory), i.e., as an accumulation of arrived items that have not been sold yet, obeys a geometric distribution restricted by the size M, and for the probability mass function, the following can be obtained:

$$p_n(\rho) = \frac{\rho^n}{\text{geom}_M(\rho)} = \frac{\rho^n}{\sum_{j=0}^{M} \rho^j}, \quad n = 0, 1, \ldots, M; \quad \rho > 0 \tag{3}$$

$$p_n(\rho) = \begin{cases} \dfrac{1-\rho}{1-\rho^{M+1}} \cdot \rho^n, & \rho \neq 1, \\ \dfrac{1}{M+1}, & \rho = 1, \end{cases} \quad n = 0, 1, \ldots, M \tag{4}$$

Therefore, for $\rho = 1$, the geometric distribution collapses into a uniform distribution across all allowed states n, $n = 0, 1, \ldots, M$. Using Eq. (4), the stationary probability of every state of the warehouse can be computed. Furthermore, by using these steady-state probabilities, vital operational measures can be extracted for a warehouse (i.e., key performance indicators for a stock manager).

The crucial physical measures can be derived for managing the stock warehouse:

(i) Stockout probability p_0: The probability that the warehouse is completely empty, meaning incoming demand cannot be immediately fulfilled (lost sales or backlogging triggers).

$$p_0(\rho) = \frac{1-\rho}{1-\rho^{M+1}}, \quad \rho > 0 \tag{5}$$

(ii) Full warehouse/procurement blocking probability p_M: The probability that the warehouse is entirely full. At this point, the supply channel is blocked, which might translate to turning away delivery trucks or pausing manufacturing lines.

$$p_M(\rho) = \frac{1-\rho}{1-\rho^{M+1}} \cdot \rho^M, \quad \rho > 0 \tag{6}$$

(iii) Average inventory level $\overline{N}$: The expected number of items in stock at any given time.

$$\bar{N}(\rho) \underline{\underline{\text{def}}} \sum_{n=0}^{M} n \cdot p_n(\rho) = \rho \cdot \frac{[\text{geom}_M(\rho)]'_\rho}{\text{geom}_M(\rho)} = \rho \cdot [\ln(\text{geom}_M(\rho))]'_\rho, \quad \rho > 0 \tag{7}$$

To resolve this equation completely as a pure function of system input parameters ρ and M, the following can be derived:

$$\overline{N}(\rho) = \begin{cases} \dfrac{\rho}{1-\rho} - \dfrac{(M+1) \cdot \rho^{M+1}}{1-\rho^{M+1}}, & \rho \neq 1 \\ M/2, & \rho = 1 \end{cases} \tag{8}$$

$$\lim_{\rho\to 0^+} \overline{N}(\rho) = 0, \quad \lim_{\rho\to +\infty} \overline{N}(\rho) = M$$

where, $\overline{N}(\rho)$ is a monotonically increasing function of the parameter ρ. It can be noted that, if $M \to \infty$, then $\overline{N}(\rho)$ is a strictly unlimited function of the parameter ρ (with $\rho < 1$).

(iv) Inventory entropy S: Staying in any of its particular states n in an equilibrium utilization, a given warehouse possesses (i.e., the variable N carries) a quantity of information $i_n(\rho)$, with possible values and its expected value, i.e., entropy $S(\rho)$, in Eqs. (9) and (10), respectively:

$$i_n(\rho) \overset{\text{def}}{=} -\ln p_n(\rho), \quad n = 0, 1, \ldots, M, \quad \rho > 0 \tag{9}$$

$$S(\rho) = E(i(\rho)) \overset{\text{def}}{=} \sum_{n=0}^{M} i_n(\rho) \cdot p_n(\rho), \quad \rho > 0 \tag{10}$$

The operational chaos or uncertainty associated with the warehouse inventory level is represented by the entropy of the probability distribution, and Eqs. (5), (9) and (10) lead to the following:

$$S(\rho) = \ln\big(\text{geom}_M(\rho)\big) - (\ln \rho) \cdot \overline{N}(\rho), \quad \rho > 0 \tag{11}$$

Eq. (11) is well known [19] and also presents the relationship between the entropy $S(\rho)$ of the warehouse, and the expected inventory level $\overline{N}(\rho)$ inside the warehouse. Combining Eq. (8) yields the final closed-form expression for the warehouse inventory entropy:

$$S(\rho) = \begin{cases} \ln\left(\dfrac{1-\rho^{M+1}}{1-\rho}\right) - (\ln \rho) \cdot \left(\dfrac{\rho}{1-\rho} - \dfrac{(M+1)\cdot\rho^{M+1}}{1-\rho^{M+1}}\right), & \rho \neq 1 \\ \ln(M+1), & \rho = 1 \end{cases} \tag{12}$$

The function $S(\rho) \to 0$, as $\rho \to 0^+$ or $\rho \to +\infty$. Furthermore, the significant maximum uncertainty point $\rho_{M,\text{max}}$, at which system entropy $S(\rho)$ has a maximum value, is the solution of the equation obtained as $S'(\rho) = 0$.

$$\sum_{n=0}^{M} \big[n - \overline{N}(\rho)\big] \cdot (\rho^n) \cdot \ln(\rho^n) = 0 \tag{13}$$

It is obvious that $\rho_{M,\text{max}} = 1$. When $\rho = 1$ (i.e., $\lambda = \mu$), the resulting entropy collapses to its absolute upper bound. It can be noted that, if $M \to \infty$, then entropy S is a strictly unlimited function of the parameter ρ (with $\rho < 1$). Eq. (12) is depicted in Figure 1. For a more detailed analysis of the mutual change of the variables: stock inventory N and its carried information i_N, please refer to the related study [25].

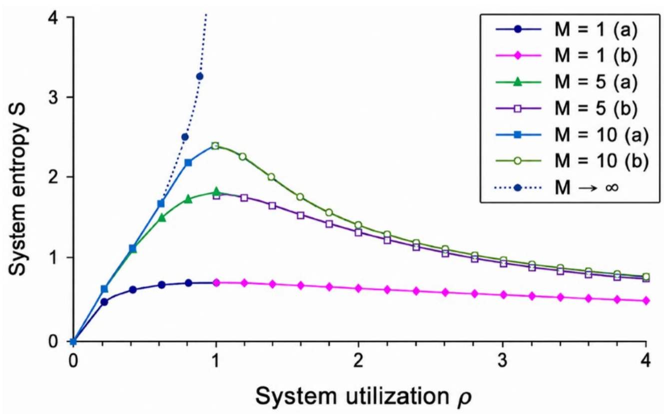

Figure 1. Entropy S vs. utilization ρ for a supply-demand warehouse

Note: $M \geq 1 \implies \rho_{M,\text{max}} = 1$; $M \to \infty \implies S$ is unlimited; (a) system is demand-driven; (b) system is supply-driven.

The utilization parameter ρ can be physically interpreted as follows:

• When $\rho = 1$, it means that the supply rate perfectly matches the demand rate. The system is a perfectly balanced system. As a managerial implication, operating at a perfect ratio of demand to capability results in maximum uncertainty. At any random evaluation check, the warehouse is equally likely to be completely blocked, completely idle, or anywhere in between.

• When $\rho < 1$, it means that demand exceeds supply. The system is demand-driven. The warehouse frequently hovers near empty ($n = 0$), and the primary operational risk is a stockout.

• When $\rho > 1$, it means that supply exceeds demand. The system is supply-driven. Inventory naturally pools up toward maximum capacity M. The primary operational risk is supply blocking or overflow.

(v) Effective supply rate λ_{eff}: Because the warehouse blocks arrivals when full, and there are no orders when empty, this is the actual rate of incoming inventory (i.e., system throughput).

$$\lambda_{\text{eff}} = \lambda \cdot \big(1 - p_M\big) = \mu \cdot \big(1 - p_0\big) \tag{14}$$

Using Eqs. (3) and (14) for the system utilization parameter ρ leads to:

$$\rho = \frac{1 - p_0(\rho)}{1 - p_M(\rho)} = \frac{1 - \dfrac{1}{\text{geom}_M(\rho)}}{1 - \dfrac{\rho^M}{\text{geom}_M(\rho)}} = \frac{\text{geom}_M(\rho) - 1}{\text{geom}_M(\rho) - \rho^M} = \frac{\text{geom}_M(\rho) - 1}{\text{geom}_{M-1}(\rho)} \tag{15}$$

Then, it is obvious that

$$p_M(\rho) \gtrless p_0(\rho) \iff \rho \gtrless 1$$

Therefore, the point $\rho_{(p_0 = p_M)} = 1$ is the break-point for a supply/demand-driven system. Comparing with Eq. (13), for this M/M/1/M system, it can be seen that $\rho_{M,\text{max}} = 1$; thus, these two points are identical.

But, in general, from Eq. (1), it follows:

$$p_M(\rho) \gtrless p_0(\rho) \iff a_M \rho^M \gtrless 1 \iff \rho \gtrless \sqrt[M]{1/a_M}$$

Furthermore, the probability that the system is neither in stockout nor blocking, i.e., it is in a real working position, is given as:

$$p_{\text{work}}(\rho) = 1 - p_0(\rho) - p_M(\rho)$$

In general, using Eq. (1) leads to:

$$p_{\text{work}}(\rho) = 1 - \frac{1}{f_M(\rho)} - \frac{a_M \rho^M}{f_M(\rho)} = \frac{\sum_{n=1}^{M-1} a_n \cdot \rho^n}{f_M(\rho)} \tag{16}$$

Furthermore, for this M/M/1/M system, using Eq. (4) leads to:

$$p_{\text{work}}(\rho) = \begin{cases} \rho \cdot \dfrac{1 - \rho^{M-1}}{1 - \rho^{M+1}}, & \rho \neq 1 \\ \dfrac{M-1}{M+1}, & \rho = 1 \end{cases} \tag{17}$$

The measures $p_0(\rho)$, $p_M(\rho)$, and $p_{\text{work}}(\rho)$ are depicted in Figure 2, with respect to the system utilization parameter ρ and system size M.

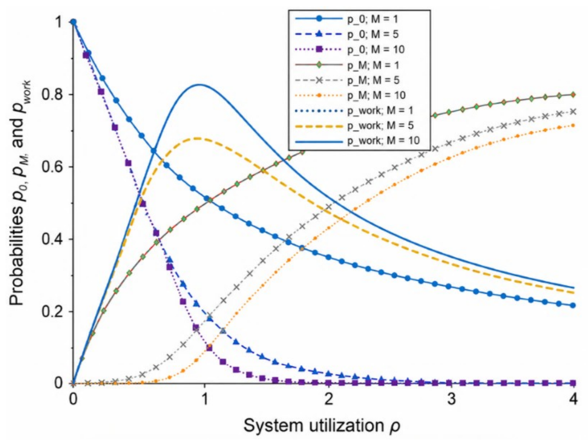

Figure 2. Probabilities p_0, p_M, and p_{work} vs. utilization ρ for a supply-demand warehouse

Note: $M \geq 1 \Rightarrow \rho_{(p_0 = p_M)} = 1$; $M \to \infty \Rightarrow p_0 \to 1 - \rho$, $p_M \to 0$, and $p_{\text{work}} \to \rho$, where $\rho < 1$.

Thus, a warehouse manager can see how far the current inventory level is from the two limit conditions (stockout and blocking) so that the stock can be regulated by changing the supply rate. It is worth noting that at the point $\rho = 1$, $p_{\text{work}}(\rho)$ has a maximum value, and as the size M increases, so $p_{\text{work}}(1)$ gets larger. In addition, as $\rho \to 0$, $p_0(\rho)$ dominates, and, as $\rho \to +\infty$, $p_M(\rho)$ dominates.

3 Managing Performance Measures

This study considers a question that commonly comes up in industry, but is not immediately obvious. The stockout probability p_0 and procurement blocking probability p_M are assessed in this study. The question is which of these two probabilities is larger in an inventory system, i.e., what is the probability that a stockout occurs before blocking $p_{\text{(out-before-block)}}$, and vice versa, the probability that blocking occurs before a stockout, $p_{\text{(block-before-out)}}$?

Therefore, this study examines the probability that the stockout occurs and the blocking does not occur, conditioned on the fact that at least one of them has occurred, which is derived as:

$$\begin{aligned} p_{\text{(out-before-block)}} &= \text{Prob}\{\text{stockout \& not(blocking)}|\text{not both}\} \\ &= \frac{\text{Prob}\{\text{stockout \& not(blocking)}\}}{\text{Prob}\{\text{not(both not)}\}} = \frac{p_0 \cdot (1-p_M)}{1-(1-p_0)\cdot(1-p_M)} \end{aligned} \tag{18}$$

Similarly, this study also examines the probability that the blocking occurs and the stockout does not occur, conditioned on the fact that at least one of them has occurred, which is derived as:

$$\begin{aligned} p_{\text{(block-before-out)}} &= \text{Prob}\{\text{blocking \& not(stockout)}|\text{not both}\} \\ &= \frac{\text{Prob}\{\text{blocking \& not(stockout)}\}}{\text{Prob}\{\text{not(both not)}\}} = \frac{p_M \cdot (1-p_0)}{1-(1-p_0)\cdot(1-p_M)} \end{aligned} \tag{19}$$

Using the complement rule in total probability for the probability that blocking and stockout are equally likely to occur, the following can be derived:

$$\begin{aligned} p_{\text{(block=out)}} &= \text{Prob}\{\text{stockout \& blocking}|\text{not both}\} \\ &= \frac{\text{Prob}\{\text{stockout \& blocking}\}}{\text{Prob}\{\text{not(both not)}\}} = \frac{p_0 \cdot p_M}{1-(1-p_0)\cdot(1-p_M)} \end{aligned} \tag{20}$$

In general, using Eq. (1) leads to:

$$p_{\text{(out-before-block)}} = \frac{p_0 \cdot (1-p_M)}{1-(1-p_0)\cdot(1-p_M)} = \frac{f_M(\rho) - a_M\rho^M}{f_M(\rho) + (f_M(\rho)-1)\cdot a_M\rho^M},\ \rho > 0$$

$$p_{\text{(block-before-out)}} = \frac{p_M \cdot (1-p_0)}{1-(1-p_0)\cdot(1-p_M)} = \frac{(f_M(\rho)-1)\cdot a_M\rho^M}{f_M(\rho) + (f_M(\rho)-1)\cdot a_M\rho^M},\ \rho > 0$$

$$p_{\text{(block = out)}} = \frac{p_0 \cdot p_M}{1-(1-p_0)\cdot(1-p_M)} = \frac{a_M\rho^M}{f_M(\rho) + (f_M(\rho)-1)\cdot a_M\rho^M},\ \rho > 0$$

Furthermore, for this M/M/1/M system, using Eq. (4) leads to:

$$p_{\text{(out-before-block)}} = \frac{p_0 \cdot (1-p_M)}{1-(1-p_0)\cdot(1-p_M)} = \begin{cases} \dfrac{(1-\rho)\cdot(1-\rho^M)}{(1-\rho^{M+1})^2 - \rho\cdot(1-\rho^M)^2}, & \rho \neq 1 \\ \dfrac{M}{2\cdot M+1}, & \rho = 1 \end{cases}$$

$$p_{\text{(block-before-out)}} = \frac{p_M \cdot (1-p_0)}{1-(1-p_0)\cdot(1-p_M)} = \begin{cases} \dfrac{(1-\rho)\cdot(1-\rho^M)\cdot\rho^{M+1}}{(1-\rho^{M+1})^2 - \rho\cdot(1-\rho^M)^2}, & \rho \neq 1 \\ \dfrac{M}{2\cdot M+1}, & \rho = 1 \end{cases}$$

$$p_{\text{(block = out)}} = \frac{p_0 \cdot p_M}{1-(1-p_0)\cdot(1-p_M)} = \begin{cases} \dfrac{(1-\rho)^2\cdot\rho^M}{(1-\rho^{M+1})^2 - \rho\cdot(1-\rho^M)^2}, & \rho \neq 1 \\ \dfrac{1}{2\cdot M+1}, & \rho = 1 \end{cases}$$

The measures $p_{\text{(out-before-block)}}(\rho)$, $p_{\text{(block-before-out)}}(\rho)$, and $p_{\text{(block = out)}}(\rho)$ are depicted in Figure 3, with respect to the system utilization parameter ρ and system size M.

Thus, the warehouse manager, for each utilization value ρ (which can be experimentally determined through measured supply rate and demand rate), can see what is more likely to happen: a stock shortage, or a procurement blocking.

It is worth noting that at the point $\rho = 1$, $p_{\text{(block = out)}}(\rho)$ has a maximum value, and as the size M increases, so $p_{\text{(block = out)}}(1)$ gets smaller. In addition, as $\rho \to 0$, $p_{\text{(out-before-block)}}(\rho)$ dominates, and, as $\rho \to +\infty$, $p_{\text{(block-before-out)}}(\rho)$ dominates.

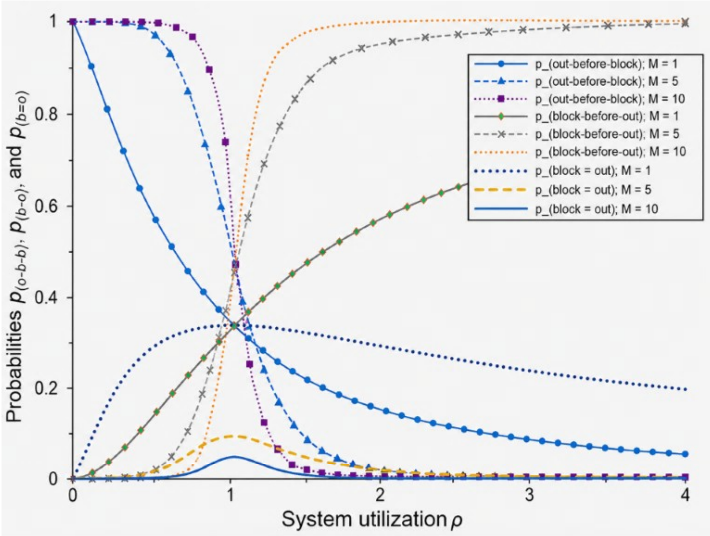

Figure 3. Probabilities $p_{(\text{out-before-block})}$, $p_{(\text{block-before-out})}$, and $p_{(\text{block }=\text{ out})}$ vs. utilization ρ for a supply-demand warehouse

Note: $M \geq 1 \implies p_{(\text{out-before-block})}(1) = p_{(\text{block-before-out})}(1) = M/(2M+1)$; and $p_{(\text{block = out})}(1) = 1/(2M+1)$. As $M \to \infty$, we have $p_{(\text{out-before-block})} \to 1$, $p_{(\text{block-before-out})} \to 0$, and $p_{(\text{block = out})} \to 0$, where $\rho < 1$.

Comparing Figure 1, Figure 2, and Figure 3, it can be seen that the maximum uncertainty point $\rho_{M,\max} = 1$ (which indicates that near this value, at any random evaluation check-in, the system is equally likely to be blocked, idle, or anywhere in between, maximizing the cognitive tracking load of the facility manager), coincides with the break-point for a supply/demand-driven system $\rho_{(p_0=p_M)} = 1$ (which indicates that near this value, the warehouse is equally far from blocking and stockout, maximizing both the probability that the system is neither in stockout nor blocked, i.e., it is in both working positions, and the probability that blocking and stockout are equally likely to occur).

3.1 Discussion

When $\rho = 1$, the warehouse is described by the uniform state probability. Then:

(i) Every single state in the birth-death process has an identical probability of occurring; $p_n = 1/(M+1)$, n = 0, 1, . . . , M. At any random point in the day, the warehouse is just as likely to be completely empty as it is to be fully maxed out or partially occupied.

(ii) The risk of stockout, p_0, is perfectly balanced against the risk of supply blocking, p_M.

(iii) The average stock inventory sits precisely at the midpoint of the available warehouse capacity ($\bar{N} = M/2$).

As a symmetric balance, considering two inverse symmetrical points of the warehouse traffic intensity (i.e., utilization parameter) ρ with respect to the point $\rho_{M,\max} = \rho_{(p_0=p_M)} = 1$ (i.e., $\rho' \cdot \rho'' = 1$), the following can be derived:

$$\bar{N}(\rho') + \bar{N}(\rho'') = M, \quad S(\rho') = S(\rho'')$$

$$p_0(\rho') = p_M(\rho''), \quad p_M(\rho') = p_0(\rho''), \quad p_{\text{work}}(\rho') = p_{\text{work}}(\rho'')$$

$$p_{(\text{out-before-block})}(\rho') = p_{(\text{block-before-out})}(\rho''), \quad p_{(\text{block-before-out})}(\rho') = p_{(\text{out-before-block})}(\rho'')$$

$$p_{(\text{block=out})}(\rho') = p_{(\text{block=out})}(\rho'')$$

3.2 Smart Logistics Center

To provide the necessary clarity for an industrial application, this study presents a structured example, illustrating how the proposed model integrates into a smart logistics center. In a smart logistics center, the primary operational challenge is balancing throughput with resource utilization under fluctuating demand. The proposed model functions as an "analytical decision engine" within the warehouse execution system. The model sits between the high-level warehouse execution system and the execution-layer hardware (like sorters, picking stations, etc.). On a regular basis such as every 10 minutes, the model pulls the real-time exact occupancy level N, aiming to calculate the empirical average inventory level $\bar{N}$ against historical benchmarks. This model can also pull the supply rate per hour λ, and the demand rate per hour μ, aiming to calculate the traffic intensity (or utilization factor) as $\rho = \lambda/\mu$. Then, all the above measures for managing the stock warehouse, described in Section 2, can be computed. Basically, the system is classified as demand-driven (i.e., $\rho < 1$) where the operational risk is a stockout, or as supply-driven (i.e., $\rho > 1$) where the operational risk is supply blocking. Furthermore, for a given traffic intensity (i.e., utilization) ρ, the model determines which event is more likely to occur: blocking or stockout.

To integrate the proposed model into a smart logistics center, the data pipeline must ensure low-latency communication between the physical assets and the decision engine. The following architecture outlines how raw operational data is transformed into actionable indicators for the dashboard:

(i) Data ingestion layer (the sensors): The pipeline begins by capturing data streams from various sources across the warehouse floor, such as real-time data from programmable logic controller tags on conveyors, sorters, and automated storage and retrieval systems; and event-based triggers from the warehouse management system.

(ii) Processing and modeling layer: A real-time database stores the current and rolling-window history of indicators, and this allows for trend analysis. This is where the proposed model resides, acting as the analytical core.

(iii) Dashboard and actuation layer: The final layer bridges the gap between insight and action. The dashboard provides the facility manager with a real-time "control room" view. Through automated feedback, the system sends instructions back to the warehouse management system, or individual hardware controllers, to adjust operational parameters (e.g., changing sortation logic or shifting labor resources) based on the calculated values.

By implementing this pipeline, the model, from a theoretical construct, becomes a dynamic controller, allowing the smart logistics center to achieve a state of "self-balancing" operations. Thus, the model generates operational indicators for daily operation that trigger automated rule adjustments, shifting the system from reactive to proactive management. By integrating these indicators, the facility manager moves away from "firefighting" (e.g., manually clearing a jam) toward prescriptive orchestration. Namely,

• Without the model: The system only responds when a warehouse reaches capacity (or stockout), causing a sudden ripple effect of delays across the entire warehouse.

• With the model: The system detects the increasing procurement blocking probability (or increasing stockout probability) and proactively throttles non-urgent inbound tasks, smoothing the flow and ensuring that high-priority orders are processed without delay.

Remark 1. Using a given sample with size m for the current stock N, if the estimation of the warehouse utilization parameter ρ is a concern, then, using Eq. (11), the entropy-based estimation of the parameter coincides with the maximum likelihood estimation (based on the average number of tasks).

Remark 2. The uncertainty has been integrated into the performance optimization of various systems. For example, the approach is applied for revenue analysis of parking lots, providing an optimal trade-off among the measures: full parking lot revenue, parking lot mean revenue, and uncertainty (i.e., entropy) of the parking lot, with respect to the parking lot utilization parameter [26]. Moreover, besides the uncertainty (i.e., risk) captured by an outside observer, the risk captured by an arriving customer and a departing customer in a queueing system has also been investigated and contrasted. It can be noted that the current paper examines solely the risk captured by an outside observer's viewpoint.

4 Conclusion and Future Research

By considering a supply-demand warehouse in this study, the current stock inventory level N can take any integer value in the range $N \in \{0, 1, \ldots, M\}$, where M is the inventory size. Then, this study analyzes the amount of information i_N carried by the random variable N, and also its average value, i.e., the entropy $S = E(i_N)$, which is the uncertainty associated with the system. Furthermore, this study derives the other measures for managing the stock warehouse, such as stockout probability, and procurement blocking probability, and evaluates which of these two probabilities is larger in an inventory system, i.e., what is the probability that a stockout occurs before a blocking, and vice versa, the probability that blocking occurs before a stockout. This study investigates all these measures with respect to the system utilization parameter ρ and system size M.

It can be seen that the maximum uncertainty point $\rho_{M,\max}$, and the break-point for a supply/demand-driven system $\rho_{(p_0=p_M)}$ are identical, and both equal one. In general, regarding the utilization parameter ρ:

(i) If $\rho = 1$, then the supply rate perfectly matches the demand rate, and the system is perfectly balanced. As a managerial implication, operating at a perfect ratio of demand to capability results in maximum uncertainty. At any random evaluation check, the warehouse is equally likely to be completely blocked, completely idle, or anywhere in between.

(ii) If $\rho < 1$, then demand exceeds supply, and the system is demand-driven. The warehouse frequently hovers near empty state 0, and the primary operational risk is a stockout.

(iii) If $\rho > 1$, then supply exceeds demand, and the system is supply-driven. Inventory naturally pools up toward maximum capacity M, and the primary operational risk is a supply blocking or overflow.

The proposed decision support engine provides a robust, analytically grounded framework for monitoring and optimizing stock inventory levels. By leveraging stochastic birth-death processes, the model facilitates prescriptive orchestration that outperforms traditional heuristic-based systems. However, it is important to situate these findings within the model's current operational assumptions. The present framework assumes a steady-state environment—specifically, constant supply and demand rates—and considers a single-warehouse architecture.

Future extensions of this work will aim to relax these assumptions in several key directions:

• Dynamic non-stationarity: Incorporating time-varying supply and demand rates to account for peak-hour surges and seasonal demand shifts.

• Multi-node orchestration: Expanding the model to support distributed logistics networks, enabling the analytical decision support engine to synchronize flow between interconnected warehouses and transit hubs.

By addressing these variables, the framework will evolve into a more versatile instrument for managing complex, globalized industrial operations, shifting from local performance tuning to global supply chain resilience.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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